

Structures, Unions and Typedef in C

[DEF] Definition to Remember

A structure (struct) is a user-defined composite data type in C that groups logically related variables of different data types under a single unified name.

1. Concept Overview & Fundamentals

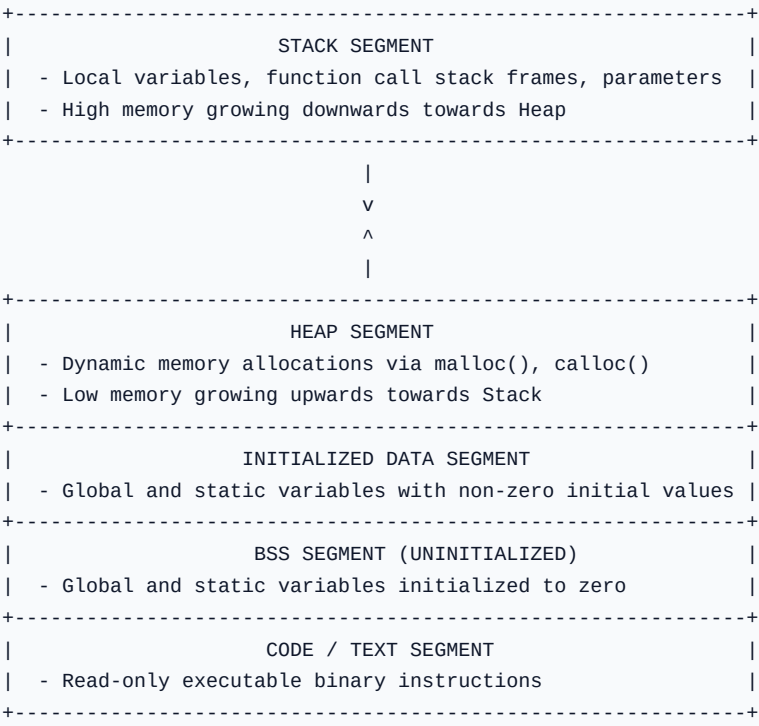
Structures, Unions and Typedef in C forms a core cornerstone of computer systems programming, hardware-level software engineering, and competitive problem solving.

In the C execution model:

* Co

2. Technical Architecture & Memory Mechanics

Memory Layout & Storage Organization



Core Architecture Breakdown

Component / Concept	Technical Specification	Operational Characteristics	Performance / Complexity
Storage Allocation	Stack vs Heap vs Data Segment	Stack is ultra-fast LIFO; Heap offers	Stack: O(1) push/pop; Heap: O(1)
Type Checking	Static Typing	Compiler verifies types during compila	Zero runtime type penalty
Addressing Model	Byte-addressable physical RAM	Pointer stores hexadecimal byte memor	Direct memory dereference (O(1))
Execution Flow	Procedural / Sequential	Controlled via jumps, branches, loops,	Deterministic instruction cycles

3. Core Syntax & Implementation Patterns

Essential Idioms & Header Declarations

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>

// Core standard implementation pattern
int main(void) {
    // Explicit initialization
    int status = 0;

    // Core logic execution
    printf("Executing C Module: Structures\n");

    return status;
}
```

4. Real-World Applications & System Engineering

* Operating System Kernels: Linux, UNIX, and Windows NT kernels utilize C for drivers, process scheduling, and memory paging.

* En

5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: book array initializer

Problem Description & Objective:

C Code Implementation:

Implem

```
/*
Initialia
differen
*/

#inclu

// Str
struct
    ch
    ch
    fl
};

int ma
/*
*
*
*
*
*
*
*
*/
```

```
st
};
in
pr
fo
}
re
}
```

Problem 2: book structure

Problem Description & Objective:

C Code Implementation:

Implem

```
/*
Create
data f
and pr
suitab
*/

#include

// Str
struct
ch
ch
fl
};

int ma
/*
*
author
*
*
*
*

st

pr
fg

pr
```

```
fg
pr
sc

pr
pr
pr
pr

re
}
```

Problem 3: car description function

Problem Description & Objective:

Implem

C Code Implementation:

```
/*
Pass a
out a
*/

#include

// Str
struct
    ch
    ch
    in
    ch
};

// Fun
void p

    pr

}

int ma
/*
*
*
*
*
*
*
*

st

};

//
```

Problem 4: car structure

Problem Description & Objective:

C Code Implementation:

```
pr  
re  
}
```

Implem

```
/*  
Define  
make,  
*/  
  
#inclu  
  
// Str  
struct  
    ch  
    ch  
    in  
    ch  
};  
  
int ma  
/*  
*  
*  
*  
*  
*  
*  
*  
  
st  
  
};  
  
pr  
pr  
pr  
pr  
pr  
  
re  
}
```

Problem 5: nested structure student books

Problem Description & Objective:

C Code Implementation:

Implem

```
/*  
Write  
also h
```

```
showin
*/

#include

// Nes
struct
    ch
    ch
};

// Mai
struct
    in
    ch
    st
};

// Fun
void c

    pr
    pr

    pr
    pr
    pr
}

int ma
/*
*

struct
*
*
*
*

st

};

di

re

}
```

Problem 6: student modify gpa

Problem Description & Objective:

C Code Implementation:

Implem

```
/*
Write
Studen
year,
*/
```

```
#include  
  
// Str  
struct  
    in  
    ch  
    in  
    fl  
};  
  
// Fun  
void m  
  
    s-  
}  
  
int ma  
/*  
*  
*  
*  
*  
modifi  
*  
*  
*  
*  
  
st  
  
};  
  
pr  
pr  
  
//  
mo  
  
pr  
pr  
re  
}
```

6. Exam Scoring Points & Memory Revision

[MUST-WRITE] Must-Write Points

1. struct definition: Groups heterogeneous data fields; accessed via dot operator (obj.field).
2. Pointer to struct: Access members using arrow operator (ptr->field == (*ptr).field).
3. Structure Padding & Memory Alignment: Compiler inserts padding bytes to align data on word boundaries.
4. typedef keyword creates intuitive aliases (e.g. typedef struct Student Student;).
5. Union vs Structure: struct allocates sum of member sizes (plus padding); union shares single memory location equal to largest member.

[RECALL] Quick Recall

Heterogeneous Grouping -> struct definition -> dot (.) / arrow (->) -> Alignment Padding -> Union Share